BLOWOUT PREVENTER TECHNICAL DOCUMENTATION

IDENTITY INFORMATION

Company_Name: Meridian Offshore Operations LLC

Field_Engineer: James R. Patterson

Field_Engineer_Id: MO-ENG-4782

Document_Version: 3.2

Machine_Name: Deepwater BOP Stack Assembly - Gulf Platform 7

Machine_Id: DW-BOP-GP7-001

Installation_Date: 2022-06-15

Number_Of_Units_In_Machine: 3

UNIT 1: ANNULAR PREVENTER (PRIMARY)

Unit_Name: Annular Preventer - Primary Seal

Unit_Id: AP-PSL-7001-A

Manufacture_Date: 2022-03-22

Manufacturer_Company_Name: Hydro-Seal Technologies Inc.

Manufacturer_Company_Id: HST-MFG-0847

Inventory_Quantity: 2 units in stock

Supplier_Information: Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 8 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 10,000 feet TVD

- **Operating_Pressure_PSI:** 5,000 PSI rated; 4,200 PSI working pressure

- **Drilling_Fluid_Compatibility:** Oil-based mud, synthetic-based mud, water-based mud (WBM)

- **Known_Error_Codes:** ERR-AP-001 (Elastomer degradation), ERR-AP-002 (Pressure seal loss), ERR-AP-003 (Poppet valve stiction)

- **Replacement_Part_Number:** HST-AP-5K-2022-REV-C

- **Maintenance_Schedule:** Pressure test every 30 days; elastomer inspection every 90 days; full overhaul every 24 months

SECTION 1 - USE CASE

- Primary emergency well control device; seals open wellbore by compressing elastomer element around drill pipe

- Provides rapid closure capability for uncontrolled well conditions (blowout scenarios)

- Maintains seal integrity across variable pipe diameters (2.375" to 5.5" OD)

- Operates in conjunction with ram preventers for redundant barrier system per API RP 53

- Critical component in IADC well control hierarchy; first line of defense in emergency response

SECTION 2 - FAILURE CAUSES

- **ERR-AP-001 (Elastomer Degradation):** Frequency 35% | Caused by thermal cycling, incompatible drilling fluid exposure, UV degradation during storage

- **ERR-AP-002 (Pressure Seal Loss):** Frequency 28% | Poppet valve wear, corrosion on sealing surfaces, inadequate maintenance intervals

- **ERR-AP-003 (Poppet Valve Stiction):** Frequency 22% | Mud accumulation in valve cavity, salt precipitation, insufficient flushing procedures

- **ERR-AP-004 (Hydraulic Line Rupture):** Frequency 10% | Abrasion from pipe movement, fatigue cracking, improper routing

- **ERR-AP-005 (Control System Malfunction):** Frequency 5% | Solenoid valve failure, electrical connector corrosion, control fluid contamination

SECTION 3 - SOLUTIONS

- **ERR-AP-001 Mitigation:** Replace elastomer element per HST service bulletin SB-2022-14; store in climate-controlled facility (60-75°F); verify fluid compatibility before deployment per API RP 54 Section 4.2

- **ERR-AP-002 Mitigation:** Conduct pressure test to 1.5x working pressure (6,300 PSI) per OSHA 1910.119(e); replace poppet valve assembly if leakage exceeds 5 drops/minute; document all test results

- **ERR-AP-003 Mitigation:** Implement daily flushing protocol with compatible spacer fluid; inspect valve cavity during 30-day maintenance; replace control fluid per manufacturer specifications

- **ERR-AP-004 Mitigation:** Reroute hydraulic lines per engineering drawing DW-BOP-GP7-HYD-001; use protective sleeving in high-movement zones; inspect for abrasion weekly

- **ERR-AP-005 Mitigation:** Test solenoid response time (must close within 30 seconds per API RP 53); replace control fluid annually; clean electrical connectors with isopropyl alcohol; verify continuity on all circuits

SECTION 4 - INVENTORY DATA

- **Current Stock:** 2 complete annular preventer assemblies; 4 elastomer element kits; 6 poppet valve cartridges

- **Reorder Threshold:** Maintain minimum 1 complete unit + 2 elastomer kits; trigger reorder when stock reaches threshold

- **Supplier:** Gulf Coast Industrial Supply (primary); backup supplier: Offshore Equipment Logistics, New Orleans LA

- **Lead Time:** 8 weeks standard; 4 weeks expedited (25% surcharge); emergency stock available at regional warehouse (2-day delivery)

UNIT 2: RAM PREVENTER (PIPE RAM)

Unit_Name: Pipe Ram Preventer - Dual Bore

Unit_Id: PR-DBR-7001-B

Manufacture_Date: 2022-04-10

Manufacturer_Company_Name: Hydro-Seal Technologies Inc.

Manufacturer_Company_Id: HST-MFG-0847

Inventory_Quantity: 1 unit in stock

Supplier_Information: Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 10 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 10,000 feet TVD

- **Operating_Pressure_PSI:** 5,000 PSI rated; 4,200 PSI working pressure

- **Drilling_Fluid_Compatibility:** Oil-based mud, synthetic-based mud, water-based mud (WBM)

- **Known_Error_Codes:** ERR-PR-001 (Ram element wear), ERR-PR-002 (Bore seal leakage), ERR-PR-003 (Accumulator pressure loss), ERR-PR-004 (Stripper element failure)

- **Replacement_Part_Number:** HST-PR-DBR-5K-2022-REV-B

- **Maintenance_Schedule:** Pressure test every 30 days; ram element inspection every 60 days; accumulator recharge every 180 days; full overhaul every 24 months

SECTION 1 - USE CASE

- Secondary emergency closure device; seals wellbore by shearing pipe or sealing against pipe OD using elastomer-faced ram blocks

- Provides positive mechanical lock for well control during emergency situations

- Dual-bore configuration allows simultaneous closure on two pipe strings or sequential closure capability

- Maintains structural integrity under full rated pressure (5,000 PSI) per API RP 53 specifications

- Integrated with accumulator system for rapid actuation independent of rig hydraulic power

SECTION 2 - FAILURE CAUSES

- **ERR-PR-001 (Ram Element Wear):** Frequency 40% | Repetitive cycling, abrasive drilling fluid particles, inadequate lubrication of ram surfaces

- **ERR-PR-002 (Bore Seal Leakage):** Frequency 25% | Elastomer degradation, corrosion pitting on bore surfaces, improper seal installation

- **ERR-PR-003 (Accumulator Pressure Loss):** Frequency 20% | Nitrogen charge leakage, bladder rupture, isolation valve malfunction

- **ERR-PR-004 (Stripper Element Failure):** Frequency 10% | Thermal cycling damage, incompatible drilling fluid, mechanical damage during pipe stripping

- **ERR-PR-005 (Hydraulic Cylinder Seal Failure):** Frequency 5% | Contaminated hydraulic fluid, rod scoring, improper maintenance

SECTION 3 - SOLUTIONS

- **ERR-PR-001 Mitigation:** Replace ram element blocks per HST service bulletin SB-2022-18; apply anti-seize compound to ram surfaces; reduce cycling frequency where operationally feasible; inspect for wear patterns every 60 days

- **ERR-PR-002 Mitigation:** Conduct pressure test to 1.5x working pressure (6,300 PSI) per OSHA 1910.119(e); replace bore seals if leakage exceeds 2 drops/minute; polish bore surfaces if pitting depth <0.015 inches

- **ERR-PR-003 Mitigation:** Recharge accumulator to 90% of system working pressure per API RP 54 Section 5.1; test isolation valve function monthly; replace bladder if nitrogen charge cannot be maintained above 80% threshold

- **ERR-PR-004 Mitigation:** Inspect stripper element for cracks/tears during 60-day maintenance; replace if damage observed; verify drilling fluid compatibility before operations; limit pipe stripping speed to <50 feet/minute

- **ERR-PR-005 Mitigation:** Change hydraulic fluid per manufacturer schedule (annually or 2,000 operating hours); inspect cylinder rod for scoring; replace seals if internal leakage exceeds 5 drops/minute; filter all hydraulic fluid to ISO 18/16/13 cleanliness

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete pipe ram assembly; 3 ram element block sets; 2 stripper element kits; 4 bore seal cartridges

- **Reorder Threshold:** Maintain minimum 1 complete unit + 2 ram element sets; trigger reorder when stock reaches threshold

- **Supplier:** Gulf Coast Industrial Supply (primary); backup: Offshore Equipment Logistics, New Orleans LA

- **Lead Time:** 10 weeks standard; 5 weeks expedited (30% surcharge); emergency stock available at regional warehouse (3-day delivery)

UNIT 3: SHEAR RAM PREVENTER

Unit_Name: Shear Ram Preventer - Pipe Cutting

Unit_Id: SR-SHP-7001-C

Manufacture_Date: 2022-05-08

Manufacturer_Company_Name: Hydro-Seal Technologies Inc.

Manufacturer_Company_Id: HST-MFG-0847

Inventory_Quantity: 1 unit in stock

Supplier_Information: Gulf Coast Industrial Supply, Houston TX | Contact: procurement@gcisupply.com | Lead Time: 12 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** 10,000 feet TVD

- **Operating_Pressure_PSI:** 5,000 PSI rated; 4,200 PSI working pressure

- **Drilling_Fluid_Compatibility:** Oil-based mud, synthetic-based mud, water-based mud (WBM)

- **Known_Error_Codes:** ERR-SR-001 (Blade dulling/chipping), ERR-SR-002 (Shear force insufficient), ERR-SR-003 (Blade misalignment), ERR-SR-004 (Seal degradation post-shear)

- **Replacement_Part_Number:** HST-SR-SHP-5K-2022-REV-A

- **Maintenance_Schedule:** Blade inspection every 30 days; shear force test every 90 days; full blade replacement every 18 months or after shear event; complete overhaul every 24 months

SECTION 1 - USE CASE

- Final emergency well control device; mechanically severs drill pipe to isolate wellbore in catastrophic blowout scenarios

- Provides absolute barrier when annular and pipe ram preventers are compromised or ineffective

- Dual-blade configuration ensures complete pipe severance across full pipe OD range (2.375" to 5.5")

- Operates at full rated pressure (5,000 PSI) per API RP 53 emergency procedures

- Integrated with BOP stack control system; actuation triggers automatic isolation of lower marine riser package (LMRP)

SECTION 2 - FAILURE CAUSES

- **ERR-SR-001 (Blade Dulling/Chipping):** Frequency 45% | Repeated contact with hard-banded drill pipe, corrosion of blade edges, inadequate blade material hardness

- **ERR-SR-002 (Shear Force Insufficient):** Frequency 30% | Hydraulic pressure loss, accumulator charge degradation, cylinder seal leakage reducing actuation force

- **ERR-SR-003 (Blade Misalignment):** Frequency 15% | Mechanical wear in pivot points, corrosion buildup on blade guides, improper maintenance assembly

- **ERR-SR-004 (Seal Degradation Post-Shear):** Frequency 7% | Thermal stress from friction during shearing, elastomer incompatibility with severed pipe material

- **ERR-SR-005 (Control System Failure):** Frequency 3% | Solenoid malfunction, electrical connector corrosion, control fluid contamination

SECTION 3 - SOLUTIONS

- **ERR-SR-001 Mitigation:** Replace shear blades per HST service bulletin SB-2022-22 after any shear event or when blade edge radius exceeds 0.030 inches; inspect blade hardness (must be 48-52 HRC); use only HST-approved replacement blades (part number HST-SR-BLADE-5K-REV-C)

- **ERR-SR-002 Mitigation:** Conduct shear force test to minimum 1,200 tons per API RP 53 Section 6.4; recharge accumulator to 90% working pressure; inspect hydraulic cylinders for internal leakage; replace cylinder seals if force drops below 1,100 tons

- **ERR-SR-003 Mitigation:** Inspect blade pivot points for corrosion/wear during 30-day maintenance; clean blade guides with soft brush and compatible solvent; verify blade alignment within ±0.010 inches per engineering drawing DW-BOP-GP7-SR-001; replace pivot pins if wear exceeds 0.005 inches

- **ERR-SR-004 Mitigation:** Allow 24-hour cooling period post-shear before pressure testing; inspect seals for thermal damage; replace all seals per HST service bulletin SB-2022-23 after shear event; verify seal material compatibility with pipe grade (API 5CT Grade X95 or equivalent)

- **ERR-SR-005 Mitigation:** Test solenoid response time (must actuate within 45 seconds per API RP 53); verify electrical continuity on all circuits; replace control fluid annually; clean connectors with isopropyl alcohol; maintain backup manual actuation capability

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete shear ram assembly; 2 shear blade sets (dual-blade configuration); 3 hydraulic cylinder seal kits; 2 accumulator bladder cartridges

- **Reorder Threshold:** Maintain minimum 1 complete unit + 1 shear blade set; trigger reorder when stock reaches threshold; maintain emergency blade stock (critical component)

- **Supplier:** Gulf Coast Industrial Supply (primary); backup: Offshore Equipment Logistics, New Orleans LA; emergency blade stock at regional warehouse

- **Lead Time:** 12 weeks standard; 6 weeks expedited (35% surcharge); emergency blade replacement available (1-week delivery); complete unit emergency stock at regional facility (5-day delivery)

COMPLIANCE CERTIFICATION

Document Prepared By: James R. Patterson, Field Engineer MO-ENG-4782

Date Prepared: 2024-01-15

Next Review Date: 2024-07-15

Certification: This document complies with OSHA 1910.119 (Process Safety Management), API RP 53 (Blowout Prevention Equipment Systems for Drilling Wells), and API RP 54 (Recommended Practices for Occupational Safety and Health Program Management in the Oil and Gas Industry).

Authorized By: Regional Operations Manager, Meridian Offshore Operations LLC

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